

CE 1201

ENGINEERING MECHANICS

Complete Backlog Preparation Guide

8 Sets

6 to Answer

210 Marks

3 Hours

Built from the materials in this chat

Primary target: CE 1201 Backlog Examination 2023. The strategy is cross-checked against the supplied regular papers (2015-2021 bundle, 2023, 2024, 2025), set-wise notes, lecture slides and solved class material. The guide focuses on what to prepare, how to solve, how to select 3 of 4 sets, and how to avoid the recurring mistakes.

Exam-focused • Bangla-assisted explanations • Formula + proof + strategy

Made by 2223033

How to use this guide

Backlog preparation should be set-based, not chapter-by-chapter.

Critical section-label warning

The 2023 Backlog paper places Statics/Equilibrium topics in Section A and Centroid/Inertia/Dynamics topics in Section B. The 2024 and 2025 regular papers follow the same broad arrangement, while the 2023 regular paper uses the opposite section order. তাই Section A/B নাম মুখস্থ না করে set/topic family চিনুন.

Exam fact	Backlog 2023
Full marks	210
Time	3 hours
Choice rule	Answer any 3 questions from each section
Questions per section	4 full questions
Marks per full question	35
Total answered sets	6 of 8

Three-layer preparation method

- Layer 1 - Select 3 primary sets in each section. এগুলো exam-ready না হওয়া পর্যন্ত extra chapter-এ যাবেন না।
- Layer 2 - Prepare the 4th set as insurance. A weak primary set can then be replaced in the exam hall.
- Layer 3 - Practice timed full questions, not isolated formulas. One full set = theory/proof + numerical + presentation.

Frequency basis used in this guide

"9/9", "6/9" etc. refer to the nine supplied regular papers analysed in this chat: 2015, 2016, 2017, 2018, 2019, 2021, 2023, 2024 and 2025. The 2020 paper was instructed to be treated as an exception, so it is not forced into the normal four-set frequency count.

Backlog 2023: 8-Set Distribution Map

Every full question is treated as one backlog set. You need 3 from each section.

SECTION A

Statics, Equilibrium, Beam, Truss & Friction Q1-Q4

Backlog Set	Full-question package	Marks	Regular-paper strength
A1 / Q1	FBD + transmissibility + two-force member + equilibrium + wheel-over-block	18+17	FBD family 9/9; wheel recurring classic
A2 / Q2	Shear force & bending moment + beam reactions/section values + pulley equilibrium	20+15	Beam family 9/9; pulley 6/9
A3 / Q3	Belt-friction proof + inverted Howe roof truss	15+20	Truss 9/9; belt proof 4/9
A4 / Q4	Slide vs tip + wedge impending motion + angle of friction vs repose	15+15+5	Friction family 9/9; slide 6/9; wedge 5/9

SECTION B

Centroid, Area M.I., Dynamics & Pappus-Guldinus Q5-Q8

Backlog Set	Full-question package	Marks	Regular-paper strength
B1 / Q5	Centroid/C.G. + applications + sector centroid + first moment of composite area	15+20	Composite centroid 9/9; theory 8/9; sector 4/9
B2 / Q6	M.I. applications + definitions + $J = I_x + I_y$ proof + polar M.I./radius numerical	5+10+20	Composite area M.I. 9/9; J proof 4/9
B3 / Q7	Linear impulse-momentum proof + inclined-body resultant/net work	16+19	Dynamics family 9/9; impulse proof 7/9
B4 / Q8	Pappus cone/torus + angle friction/repose + impulse numerical	13+8+14	Pappus 3/9 + high-overlap theory/dynamics

Most important insight

Backlog does not introduce a completely new syllabus. It recombines the same recurring families. The safest strategy is therefore: master the high-frequency family + the exact 2023 backlog version + one recent regular variant (2024/2025).

Priority & 3-out-of-4 Selection Strategy

Priority combines repetition, predictability, time risk and topic overlap.

Section A set	Repetition	Time risk	Default role	Recommended if...
A2 Beam + Pulley	Very High	Medium	Primary	You want the most stable repeat pattern
A4 Friction + Wedge	Very High	Medium	Primary	You can draw correct friction directions/FBD
A1 FBD + Wheel	High	Low-Medium	Primary / Backup	You prefer theory + one geometry equilibrium problem
A3 Belt + Truss	Very High	Higher	Primary / Backup	You are comfortable with method of joints/sections

Section A - two valid selection profiles

Profile 1 (safer for most students): A1 + A2 + A4, keep A3 as backup. Profile 2 (truss-strong): A2 + A3 + A4, keep A1 as backup. Do not skip A3 completely because truss is one of the strongest recurring families.

Section B set	Repetition	Time risk	Default role	Why
B1 Centroid	Very High	Low-Medium	Primary	Concept + standard sector + composite first moment
B2 Area M.I.	Very High	Medium	Primary	Highly formulaic and repeated
B3 Impulse + Work	Very High	Medium	Primary	Proof + one direct inclined-body numerical
B4 Pappus + Impulse	Medium-High	Low-Medium	Insurance	Friction and impulse overlap with already-prepared topics

Section B default

Prepare B1 + B2 + B3 as the main three. Prepare B4 as a low-marginal-cost backup: angle of friction is already needed in A4, impulse is already needed in B3, so only Pappus needs substantial extra preparation.

A1 / Q1

FBD + Equilibrium + Wheel Over Block 35 marks

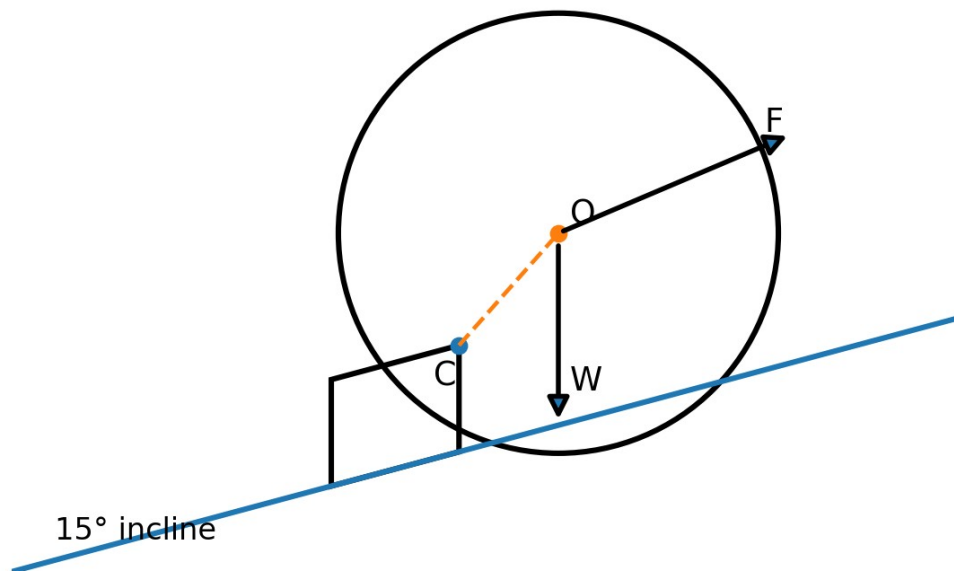
Part	Backlog 2023 question anatomy	Marks
(a)	Define transmissibility of force, two-force member, free-body diagram and equilibrium; draw FBDs of the given structure.	18
(b)	Wheel is on the point of rolling over a block; $W = 1000$ lb. Find the magnitude and sense of the least force F .	17

Why it matters

FBD/equilibrium modelling is a 9/9 broad family. The set-wise note itself tags FBD in 2015, 2016, 2017, 2018, 2021, 2022 and two-force member in 2016, 2017, 2018, 2021, 2022. The wheel-over-block family is tagged as a recurring 2015/16/17/21 pattern.

Theory answers you must be able to write

- Transmissibility: a force may be moved anywhere along the same line of action without changing the external effect on a rigid body, provided magnitude and direction remain unchanged.
- Two-force member: when only two forces act on an equilibrium member, they must be equal, opposite and collinear.
- FBD: isolate the body/member and show all external forces, weights, applied loads, support reactions, cable tensions and contact reactions.
- Equilibrium for a coplanar rigid body: $\Sigma F_x = 0$, $\Sigma F_y = 0$ and $\Sigma M = 0$.

Wheel over a block: impending rotation about corner C

At impending rotation the wheel pivots about the block corner; take moments about that corner.

Anchor answer check for the recurring wheel family

At impending rolling over corner C: reaction at the original ground contact becomes zero.

Take $\Sigma M_C = 0$ to eliminate the unknown reaction at C.

For the recurring scaled geometry in the supplied note: $\theta \approx 33.19^\circ$, $F_{\min} \approx 547.42$ lb.

Bangla shortcut

Wheel over block: "Contact বদলালে pivot বদলায়." Block corner-কে pivot ধরুন → geometry থেকে OA direction বের করুন → moment about corner → least force-এর জন্য denominator সর্বোচ্চ ($\sin = 1$) করুন.

Practice ladder

Order	Paper	What to practice
1	2017 Q1	Definitions + FBD + wheel-over-block least force/reaction
2	2015 Q1	FBD + wheel-over-block
3	2021 Q1	Near-repeat of definitions/FBD/wheel
4	2025 Q1	Recent FBD + inclined-cylinder equilibrium variant

Common errors

Drawing internal forces on the wrong isolated body; forgetting that a smooth contact reaction is normal to the surface; keeping the old contact reaction after lift-off; taking moment arm as a slanted distance rather than perpendicular distance.

A2 / Q2

Shear Force & Bending Moment + Pulley 35 marks

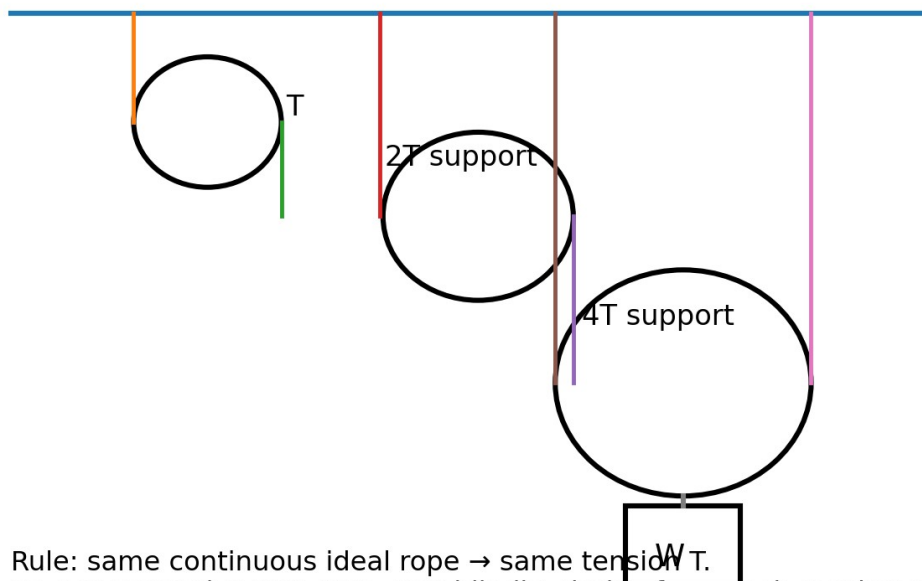
Part	Backlog 2023 question anatomy	Marks
(a)	Define shear force and bending moment. For the loaded beam, find support reactions and the shear force and bending moment at sections (a)-(a) and (b)-(b).	20
(b)	For the frictionless pulley system, determine W for equilibrium.	15

Why it is a top-priority set

Beam reactions + S.F./B.M. is a 9/9 numerical family in the analysed regular papers; the definitions appear in 8/9. Pulley equilibrium appears in 6/9. This is one of the most bankable full-question packages.

Beam solution algorithm - always in this order

- Convert distributed loads to equivalent point loads: UDL wL at its midpoint; triangular load $\frac{1}{2}wL$ at $L/3$ from the heavy end.
- Resolve inclined loads into horizontal and vertical components before equilibrium.
- Find support reactions using whole-beam $\Sigma F_x=0$, $\Sigma F_y=0$, $\Sigma M=0$.
- Cut at the requested section. Use only the left or right segment - whichever has fewer loads.
- Compute shear V from force equilibrium and bending moment M from moment equilibrium. State the sign convention you used.

Formula / relation to remember**UDL resultant = wL** **Triangular-load resultant = $\frac{1}{2} w_{\text{max}} L$** **SF at a cut = algebraic sum of transverse forces on one side****BM at a cut = algebraic sum of moments of external forces about the cut****Pulley equilibrium: tension propagation concept**

Rule: same continuous ideal rope $\rightarrow$ same tension T .
Do not memorize $W/2$, $W/4$, $W/8$ blindly; derive from each moving pulley FBD.

Pulley shortcut is conceptual, not a memorized division table.

Exact backlog-family answer check

The supplied pulley note contains the same 3 ft - 2 ft - 1 ft beam with 200 lb and 400 lb loads. Following the rope-tension ladder gives a beam pull of $W/8$; $\Sigma M_A = 0$ gives $W = 4800$ lb.

Practice ladder

Order	Paper	What to practice
1	2025 Q2	Very recent beam with mixed loads + cable companion
2	2024 Q2	Beam + cable, same full-set discipline
3	2023 Q6	Beam + pulley in one question
4	2018/2019 Q6	Classic beam reaction + section S.F./B.M. + pulley

Common errors

For a triangular load, placing the resultant at the wrong end; including a pure couple in ΣF ; omitting the couple from ΣM ; assuming every pulley halves weight automatically; using different tensions on the same ideal continuous rope.

A3 / Q3

Belt Friction + Inverted Howe Roof Truss 35 marks

Part	Backlog 2023 question anatomy	Marks
(a)	For belt friction, prove $T_1 = T_2 e^{f\theta}$, with usual notation.	15
(b)	Determine the forces in the members located to the right of DE for the inverted Howe roof truss.	20

Priority logic

Truss numerical appears as a full-question family in all 9 analysed regular papers. Belt-friction derivation appears explicitly in 4/9 and is short, predictable theory. The risk is time: truss arithmetic can be longer than A1/A4.

Belt-friction proof - the 6-line structure

- Take a small belt element subtending $d\theta$. Tension increases from T to $T+dT$ on the tight side.
- For the small element, radial equilibrium gives $dN \approx T d\theta$.
- At limiting friction, $dF = f dN = fT d\theta$.
- Tangential equilibrium gives $dT = dF = fT d\theta$.
- Integrate: $\int (dT/T) = f \int d\theta$ from T_2 to T_1 and 0 to θ .
- Therefore $\ln(T_1/T_2) = f\theta \rightarrow T_1 = T_2 e^{f\theta}$. θ must be in radians.

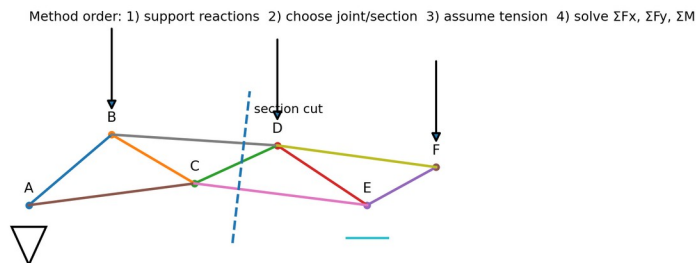
Formula / relation to remember

$$T_1 / T_2 = e^{f\theta}$$

θ in radians

T_1 = tight-side tension; T_2 = slack-side tension

Truss strategy: joints for all members, sections for selected members

**Truss shortcut**

Whole-truss reactions first. If the question asks only members to the right of DE, a section cut that crosses at most three unknown members can be much faster than solving every joint. Assume unknown member forces in tension; a negative result means compression.

Practice & error focus

Practice order: 2016 Q3 (inverted Howe - closest family) → 2024 Q3 (belt + truss) → 2017 Q4 (selected members) → 2025 Q3 (recent full truss). Avoid starting at a joint with too many unknowns; resolve inclined members; find support reactions first; report T/C; and use θ in radians in $e^{f\theta}$.

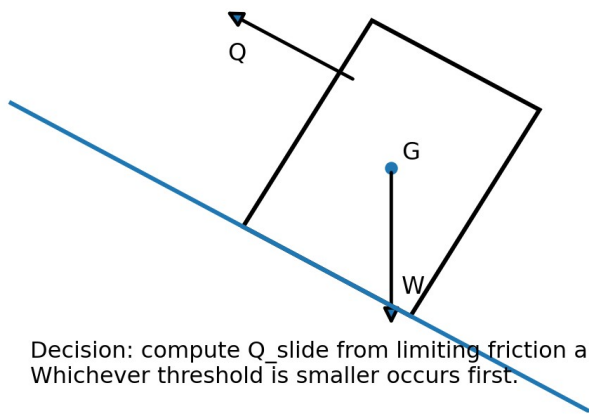
A4 / Q4

Friction: Slide vs Tip + Wedge + Angle of Repose 35 marks

Part	Backlog 2023 question anatomy	Marks
(a)	Tall slender body on a 30° incline: $W=105$ lb, $f=0.3$. With increasing Q , determine whether it slides or tips first.	15
(b)	Wedge B between fixed A and movable C: $W_A=200$ lb, $W_B=500$ lb, $W_C=5000$ lb, $\theta=7.5^\circ$, $R=13000$ lb, $f=1/3$. Find Q for impending motion of C.	15
(c)	Distinguish between angle of friction and angle of repose.	5

Why it is a primary set

The broad friction/wedge family appears in all 9 analysed regular papers. Slide-vs-tip appears in 6/9, wedge numerical in 5/9, and angle of friction/repose in 6/9. 2024 and 2025 papers are especially close to this structure.

Sliding vs overturning: compare two threshold forces

Decision: compute Q_{slide} from limiting friction and Q_{tip} from moment about downhill edge. Whichever threshold is smaller occurs first.

Core friction relations

Limiting friction: $F = \mu N$

Angle of friction: $\tan \phi = \mu$

At impending sliding: solve force equilibrium with $F=\mu N$

At impending tipping: take moments about the downhill edge; reaction passes through pivot

Bangla shortcut

“যেদিকে চলতে চায়, friction তার উল্টো।” Slide-vs-tip-এ দুটি Q বের করুন: Q_{slide} এবং Q_{tip} . যেটা কম, সেটাই আগে ঘটবে. Wedge-এ প্রতিটি contact reaction normal থেকে friction angle ϕ পরিমাণ ঘুরবে, impending motion-এর বিরোধী দিকে.

Angle of friction vs angle of repose - 5-mark answer

Point	Angle of friction, ϕ	Angle of repose, α
Definition	Angle between resultant reaction and normal at limiting friction	Maximum plane inclination at which body is just about to slide
Relation	$\tan \phi = \mu$	$\tan \alpha = \mu$

Point	Angle of friction, φ	Angle of repose, α
At limiting condition	$\varphi = \tan^{-1} \mu$	$\alpha = \tan^{-1} \mu$
Conclusion	For the same surfaces, $\varphi = \alpha$ at limiting equilibrium	Same numerical value at impending slide

Practice ladder

Order	Paper	What to practice
1	2025 Q4	Angle/repose + slide/tip + wedge - near-perfect backlog rehearsal
2	2024 Q4	Angle/repose + pulley-friction + wedge
3	2016 Q4	Laws of friction + slide/tip + wedge
4	2015/2021 friction set	Belt proof + slide/tip

Common errors

Using $F = \mu N$ even when the body is not at limiting/kinetic condition; choosing friction direction from memory instead of impending motion; forgetting the tipping pivot; drawing wedge reactions normal to surfaces even when friction is present.

Section A: One-Page Revision Checklist

Goal: make at least three full sets exam-ready and keep one backup.

Set	Minimum mastery before exam	Answer-writing trigger
A1	4 definitions + FBD rules + wheel pivot geometry	"Isolate → reactions → pivot corner → moment"
A2	Load conversion + reactions + section V/M + pulley tension propagation	"Load → reaction → cut → $\Sigma F/\Sigma M$ "
A3	Belt proof + truss reactions + joints/sections + T/C	"Reaction first → easiest joint/section"
A4	Friction direction + slide/tip thresholds + wedge reaction angles	"Impending motion first → friction opposite"

Best use of the last 30 minutes of Section A study

Do one FBD, one beam cut, one pulley tension ladder, one truss section, and one slide-vs-tip comparison without looking at notes. If any one cannot be started within 60 seconds, that set is not exam-ready.

B1 / Q5

Centroid, Center of Gravity & First Moment 35 marks

Part	Backlog 2023 question anatomy	Marks
(a)	Explain centroid and center of gravity, state real-life applications of centroid, and locate the centroid of a circular sector of radius r subtending angle 2β .	15
(b)	Determine the first moment of area about X and Y axes for the given shaded composite area.	20

Why it is the safest Section B set

Composite centroid/C.G. is a 9/9 family and centroid/C.G./C.O.M. theory appears in 8/9 analysed regular papers. The prepared centroid note tags "Define Centroid" across 2015-2022 and applications repeatedly.

Concepts in Bangla

- Centroid (জ্যামিতিক কেন্দ্র): area/line/volume distribution-এর balance point; material weight লাগবে না.
- Center of Gravity (ভারকেন্দ্র): body-এর total weight যেন একটি point-এ concentrated - সেই point.
- Uniform density and uniform gravitational field হলে C.G./C.M. and geometric centroid can coincide for a homogeneous body.

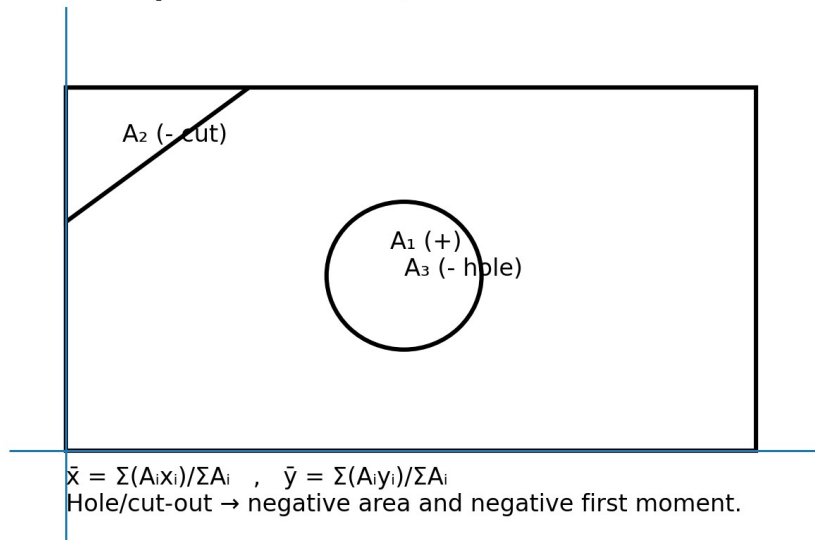
Formula / relation to remember

First moment about x-axis: $Q_x = \int y \, dA = A\bar{y}$

First moment about y-axis: $Q_y = \int x \, dA = A\bar{x}$

Composite: $\bar{x} = \Sigma(A_i x_i) / \Sigma A_i$, $\bar{y} = \Sigma(A_i y_i) / \Sigma A_i$

Circular sector (angle 2β): centroid distance = $2r \sin\beta / (3\beta)$, β in radians

Composite centroid / first moment of area**Composite-area rule**

Hole/cut-out = negative area. Keep one consistent reference axis. Make a table: shape, A_i , x_i , y_i , $A_i x_i$, $A_i y_i$. First moment is not the same as second moment of area.

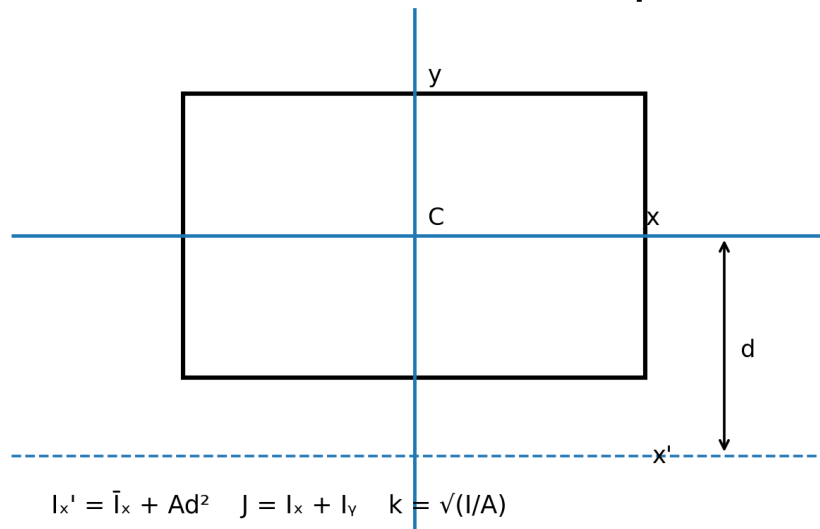
B2 / Q6

Area Moment of Inertia, Polar M.I. & Radius of Gyration **35 marks**

Part	Backlog 2023 question anatomy	Marks
(a)	State real-life applications of moment of inertia of area in construction engineering.	5
(b)	Define moment of inertia, polar moment of inertia and radius of gyration. Prove $J = I_x + I_y$.	10
(c)	Calculate polar moment of inertia and radius of gyration with respect to X and Y axes of the shown area.	20

Why it is a primary set

Composite area moment of inertia is a 9/9 family. M.I. definitions/concepts appear in 7/9 and $J = I_x + I_y$ is an explicit 4/9 derivation. This is high-mark, formula-controlled content.

Area moment of inertia: centroidal and parallel axes**Formula / relation to remember**

$$I_x = \int y^2 dA$$

$$I_y = \int x^2 dA$$

$$J_O = \int r^2 dA = I_x + I_y$$

$$\text{Parallel-axis theorem: } I = \bar{I} + Ad^2$$

$$\text{Radius of gyration: } k = \sqrt{I/A}$$

Composite/hole: locate centroid as needed; subtract hole terms ($\bar{I} + Ad^2$)

Unit: $I, J \rightarrow \text{length}^4$; $k \rightarrow \text{length}$

 $J = I_x + I_y$ proof skeleton

- For an elemental area dA at (x,y) , $r^2 = x^2 + y^2$.
- Polar M.I.: $J = \int r^2 dA = \int (x^2 + y^2) dA$.
- Separate the integral: $J = \int x^2 dA + \int y^2 dA$.
- Recognize $\int x^2 dA = I_y$ and $\int y^2 dA = I_x$.
- Therefore $J = I_x + I_y$.

B3 / Q7

Linear Impulse-Momentum + Work on Inclined Plane 35 marks

Part	Backlog 2023 question anatomy	Marks
(a)	State the principle of linear impulse and momentum and prove $R\Delta t = \Delta mv$.	16
(b)	$W=200$ lb, $\alpha=35^\circ$, $Q=125$ lb, $\beta=25^\circ$, $f=1/3$; body moves 12 ft up the incline. Find resultant force and net work.	19

Why it is a primary set

The dynamics full-question family appears in all 9 analysed regular papers. Linear impulse-momentum proof appears in 7/9, making it one of the most repeated proofs in the entire course.

Impulse-momentum proof skeleton

- From Newton's second law: $R = ma = m(dv/dt)$ for constant mass.
- Multiply by dt : $R dt = m dv$.
- Integrate over $t_1 \rightarrow t_2$ and $v_1 \rightarrow v_2$: $\int R dt = m(v_2 - v_1)$.
- For constant resultant R : $R\Delta t = m v_2 - m v_1 = \Delta(mv)$.
- Interpretation: impulse = change in linear momentum.

Formula / relation to remember

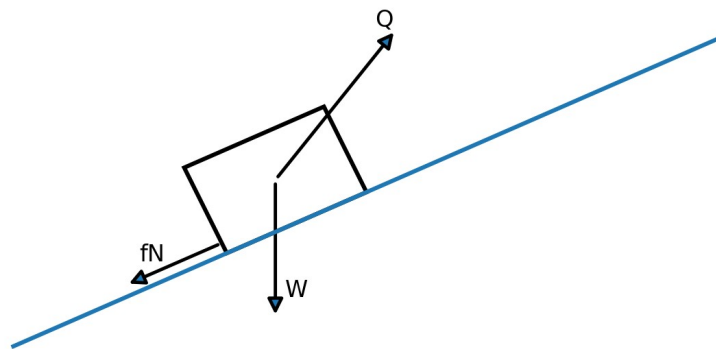
Impulse: $I = \int F dt$; for constant F , $I = F\Delta t$

Momentum: $p = mv$

FPS mass: $m = W/g$ (slug)

Net impulse = resultant force $\times$ time only when the resultant is constant

Work-energy / net-work on an inclined plane



Along displacement: + work; opposite displacement: - work.
 $N = W \cos \alpha - Q \sin \beta$ (when Q lifts the block from the plane).

Exact backlog Q7(b) answer check

Using the force decomposition consistent with the supplied work-energy class note: $N = W \cos 35^\circ - Q \sin 25^\circ \approx 111.00$ lb; friction $= (1/3)N \approx 37.00$ lb. Along the plane, $R = Q \cos 25^\circ - W \sin 35^\circ - \text{friction} \approx -38.43$ lb. So the resultant is 38.43 lb down the plane and W_{net} over 12 ft upward ≈ -461.1 ft-lb. The negative work means the resultant opposes the upward displacement. Practice next: 2023 Q4 $\rightarrow$ 2018 Q4 $\rightarrow$ 2024 Q7 $\rightarrow$ 2017 Q8.

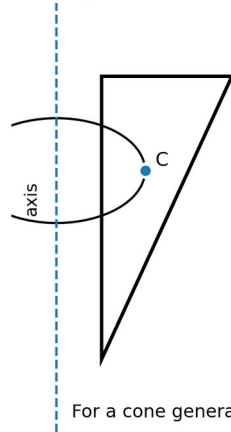
B4 / Q8

Pappus-Guldinus + Friction Theory + Impulse Numerical 35 marks

Part	Backlog 2023 question anatomy	Marks
(a)	Use Pappus-Guldinus to determine volume and paint required for inside/outside surfaces of the shown cone; the wording also references a doughnut/torus. One gallon covers 300 ft ² .	13
(b)	Distinguish between angle of friction and angle of repose.	8
(c)	90 lb body on horizontal plane, $f=0.25$, forces Q and F applied; find individual horizontal impulses during 2.5 s and net impulse.	14

Why this is a smart backup set

Pappus itself is less frequent (3/9), but 22 of the 35 marks reuse topics already needed elsewhere: angle of friction/repose from A4 and impulse from B3. That makes B4 a low-cost insurance set.

Pappus-Guldinus: centroid travels a circular pathSurface theorem: $S = L(2\pi\bar{r})$ Volume theorem: $V = A(2\pi\bar{r})$ Memory: generating quantity $\times$ centroid travel distanceFor a cone generated by a right triangle: $A = \frac{1}{2}rh$ and $\bar{r} = r/3$.**Formula / relation to remember****Surface of revolution: $S = L(2\pi\bar{r})$** **Volume of revolution: $V = A(2\pi\bar{r})$** **Torus volume: $V = 2\pi^2 a^2 b$** **Torus surface area: $S = 4\pi^2 ab$** **Cone answer check from the supplied Pappus slide**

For $r=3$ ft and $h=10$ ft: slant length $L=\sqrt{10^2+3^2}=10.4403$ ft. The generating line centroid is 1.5 ft from the axis, so one lateral surface $S \approx 98.40$ ft². Inside + outside ≈ 196.80 ft², paint $\approx 196.80/300=0.656$ gallon. Using the area theorem, cone volume ≈ 94.25 ft³. Before applying Pappus, always identify the generating curve/area and its centroid travel distance; the rotation axis must satisfy the theorem conditions.

Impulse numerical algorithm

- Resolve Q and F into horizontal and vertical components.
- Use vertical equilibrium to find N if the block stays in contact with the horizontal plane.
- Kinetic friction magnitude = fN and direction opposite motion.
- Individual horizontal impulse of each applied force = $F_x \Delta t$.
- Net impulse = $(\Sigma F_x)\Delta t$, including friction. Use mass only if converting impulse to velocity change.

Cross-Topic Leverage: Study Once, Use Twice

Backlog rewards overlap. Use it deliberately.

Study item	Where it scores	Why it is efficient
Angle of friction vs repose	A4(c) + B4(b)	Same theory can appear in both sections
Impulse-momentum	B3(a) + B4(c)	Proof + numerical share the same framework
Equilibrium / FBD	A1, A2, A3, A4	Every statics numerical begins with correct modelling
Centroid / first moment	B1 + prerequisite for B2 composite M.I.	Centroid location often needed before parallel-axis M.I.
Friction direction	A4 + B3 work + B4 impulse	One sign error can reverse several answers
Pappus + centroid	B4 + centroid understanding	Pappus is literally “generating quantity × centroid travel”

Backlog efficiency principle

Do not count chapters; count transferable procedures. Correct FBD, sign convention, centroid table, and force resolution are reusable skills across multiple sets.

Backlog Theory Bank

Write these in clean, compact exam language.

SECTION A THEORY

High-yield definitions and short notes

Term	Exam-ready answer
Free Body Diagram	Isolated-body sketch showing all external forces/reactions; remove surroundings and replace constraints by reactions.
Transmissibility	For a rigid body, moving a force along its line of action does not change the external effect if magnitude and direction are unchanged.
Two-force member	Only two external forces; for equilibrium they are equal, opposite and collinear.
Equilibrium	Resultant force and resultant moment are zero: $\Sigma F_x=0$, $\Sigma F_y=0$, $\Sigma M=0$.
Shear Force	Algebraic sum of transverse external forces on either side of a section.
Bending Moment	Algebraic sum of moments of external forces about a section, using one side.
Angle of friction	Angle between resultant contact reaction and normal reaction at limiting friction; $\tan\phi=\mu$.
Angle of repose	Maximum inclination of a plane at which a body is just about to slide; $\tan\alpha=\mu$; at limiting equilibrium $\alpha=\phi$.

SECTION B THEORY

High-yield concepts

Term	Exam-ready answer
Centroid	Geometric balance point of a line/area/volume; depends on geometry, not weight.
Center of Gravity	Point through which the resultant weight of a body acts.
Moment of inertia of area	Second moment of area about an axis; $I=\int \text{distance}^2 dA$; geometric measure of distribution of area.
Polar M.I.	Second moment about a perpendicular axis through a point; $J=\int r^2 dA=I_x+I_y$.
Radius of gyration	Equivalent distance k at which the entire area could be concentrated to give the same M.I.; $k=\sqrt{I/A}$.
Linear impulse	Time integral of force; equals change in linear momentum.
Work	Force component along displacement multiplied/integrated over displacement; sign depends on direction relative to displacement.
Pappus-Guldinus	Surface/volume of revolution equals generating length/area multiplied by the distance travelled by its centroid.

Proof / Derivation Bank

Memorize the logic, not just the final line.

Priority	Proof	Backlog location	Core chain
1	Linear impulse-momentum	B3 Q7(a)	$R = m \, dv/dt \rightarrow R \, dt = m \, dv \rightarrow \text{integrate} \rightarrow R \Delta t = \Delta(mv)$
2	Belt friction	A3 Q3(a)	$dN \approx T \, d\theta \rightarrow dF = fT \, d\theta \rightarrow dT/T = f \, d\theta \rightarrow \text{integrate}$
3	Polar relation $J = I_x + I_y$	B2 Q6(b)	$r^2 = x^2 + y^2 \rightarrow \text{integrate} \rightarrow J = I_y + I_x$
4	Centroid of circular sector	B1 Q5(a)	Use area integration or memorized sector-centroid derivation; β in radians
5	Pappus theorem application	B4 Q8(a)	Identify generator, centroid radius, travel $2\pi\bar{r}$; then $S = L \cdot \text{travel}$ or $V = A \cdot \text{travel}$

Proof-writing format for full marks

State $\rightarrow$ small diagram/element $\rightarrow$ governing equilibrium/Newton/geometric relation $\rightarrow$ integration with limits $\rightarrow$ boxed final equation $\rightarrow$ define symbols/units. Do not jump directly from the first equation to the final result.

One-line memory hooks

- Impulse: "Force $\times$ time changes momentum."
- Belt: "Small element $\rightarrow$ normal from curvature $\rightarrow$ friction changes tension."
- Polar M.I.: " $r^2 = x^2 + y^2$, so $J = I_x + I_y$."
- Pappus: "Generating quantity $\times$ centroid travel."

Backlog Formula Bank - Section A

Keep this page beside you during practice.

A1

Equilibrium & Wheel

Formula / relation to remember

$$\Sigma F_x = 0, \Sigma F_y = 0, \Sigma M = 0$$

Smooth contact → reaction normal to surface

Two-force member → equal, opposite, collinear forces

Wheel at impending climb → take moment about block corner

A2

Beam & Pulley

Formula / relation to remember

$$\text{UDL: } R = wL \text{ at } L/2$$

Triangular load: $R = \frac{1}{2}wL$ at $L/3$ from heavy end

At section: V from ΣF ; M from ΣM

Ideal same rope → same T ; movable pulley equilibrium from number/direction of supporting tension segments

A3

Belt & Truss

Formula / relation to remember

$$T_1 = T_2 e^{(f\theta)}, \theta \text{ in radians}$$

Truss whole body: $\Sigma F_x=0, \Sigma F_y=0, \Sigma M=0$

Joint: $\Sigma F_x=0, \Sigma F_y=0$

Section: $\Sigma F_x=0, \Sigma F_y=0, \Sigma M=0$; assume cut-member forces in tension

A4

Friction & Wedge

Formula / relation to remember

Limiting friction $F = \mu N$

$\tan \phi = \mu$; angle of repose $\alpha = \tan^{-1} \mu$

Slide threshold: force equilibrium + $F = \mu N$

Tip threshold: moment about edge = 0

Wedge contact resultant is inclined by ϕ from the normal, opposite impending relative motion

Backlog Formula Bank - Section B

Core equations and units.

B1
Centroid & First Moment

Formula / relation to remember

$$Q_x = A\bar{y} ; Q_y = A\bar{x}$$

$$\bar{x} = \Sigma A_i x_i / \Sigma A_i ; \bar{y} = \Sigma A_i y_i / \Sigma A_i$$

Hole/cut-out $\rightarrow A < 0$

$$\text{Sector } 2\beta: \bar{r} = 2r \sin\beta / (3\beta), \beta \text{ in radians}$$

B2
Area M.I.

Formula / relation to remember

$$I_x = \int y^2 dA ; I_y = \int x^2 dA$$

$$J = I_x + I_y$$

$$I = \bar{I} + Ad^2$$

$$k_x = \sqrt{I_x/A}, k_y = \sqrt{I_y/A}$$

$$\text{Rectangle centroidal: } I_x = bh^3/12, I_y = hb^3/12$$

B3
Impulse & Work

Formula / relation to remember

$$\text{Impulse} = \int F dt ; \text{constant } F \rightarrow F\Delta t$$

$$R\Delta t = \Delta(mv)$$

$$m = W/g \text{ in FPS}$$

$$\text{Work: } U = \int F \cdot ds$$

$$\text{Net work} = \Delta KE = \frac{1}{2}m(v_2^2 - v_1^2)$$

$$\text{Incline: } N = W \cos\alpha - Q \sin\beta \text{ when } Q \text{ acts above the plane}$$

B4
Pappus

Formula / relation to remember

$$S = L(2\pi\bar{r})$$

$$V = A(2\pi\bar{r})$$

$$\text{Torus } V = 2\pi^2 a^2 b$$

$$\text{Torus } S = 4\pi^2 ab$$

$$\text{Cone lateral generator } L = \sqrt{h^2 + r^2}$$

Regular Question Bank → Backlog Practice Map

Use recent papers first, then older near-repeats.

Backlog set	Best matching regular practice	Reason
A1	2017 Q1 → 2015 Q1 → 2021 Q1 → 2025 Q1	Definitions/FBD + wheel or contact equilibrium
A2	2025 Q2 → 2024 Q2 → 2023 Q6 → 2018/19 Q6	Beam full solution and pulley/cable companions
A3	2016 Q3 → 2024 Q3 → 2017 Q4 → 2025 Q3	Inverted Howe / selected truss members + belt proof
A4	2025 Q4 → 2024 Q4 → 2016 Q4 → 2015/21 friction set	Near-identical friction/wedge architecture
B1	2025 Q5 → 2023 Q1 → 2017 Q5 → 2024 Q5	Sector + theory + composite centroid
B2	2025 Q6 → 2023 Q2 → 2019 Q2 → 2017 Q6	Composite area M.I. + J proof
B3	2023 Q4 → 2018 Q4 → 2024 Q7 → 2017 Q8	Impulse proof + inclined work / impulse numerical
B4	2015 Q5 + 2018 Q3 + 2023 backlog Q8	Pappus examples; reuse A4/B3 for friction/impulse

Practice rule

For each primary set, solve at least 3 full-question variants: (1) backlog exact, (2) newest matching regular, (3) one older near-repeat. Mark your own script using a checklist: setup, formula, substitution, unit, final direction/sign.

14-Day Backlog Plan

A complete but realistic set-based schedule.

Day	Focus	Minimum output
Day 1	A1 theory + FBD	Write 4 definitions from memory; draw 4 FBDs.
Day 2	A1 wheel + A2 load conversion	Solve recurring wheel family; practice UDL/UVL resultants.
Day 3	A2 beam	One full backlog beam + one 2024/25 beam timed.
Day 4	A2 pulley + recap	Solve backlog pulley to W=4800 lb; redraw tension ladder without notes.
Day 5	A3 belt proof + truss basics	Write proof twice; support reactions + easy joints.
Day 6	A3 truss	Inverted Howe / method of sections practice; label T/C.
Day 7	A4 friction	Angle/repose + slide/tip + friction directions.
Day 8	A4 wedge + Section A mock	Wedge FBDs; then one timed 3-set Section A simulation.
Day 9	B1 centroid	Sector formula + composite first moment table.
Day 10	B2 M.I.	J proof + parallel-axis + composite M.I.
Day 11	B3 impulse	Proof + impulse numerical; unit conversion W/g.
Day 12	B3 work + B4 Pappus	Backlog Q7(b) + cone/torus Pappus.
Day 13	Section B mock + backup set	Timed B1+B2+B3; then repair B4 weak spots.
Day 14	Full 3-hour mock + error log	Choose 3+3 exactly as exam; final formula/proof revision only after review.

7-Day & 48-Hour Rescue Plans

Use these only if the full 14-day plan is not possible.

7-DAY PLAN

One major block per day + short revision block

Day	Main work
1	A1 + A2 theory/FBD + one full beam
2	A2 pulley + A4 friction/slide-tip
3	A3 belt/truss + A4 wedge
4	B1 centroid + B2 J proof/M.I.
5	B3 impulse/work + B4 Pappus
6	Two half-mocks: Section A 3 sets + Section B 3 sets
7	Formula/proof recall + weakest-set repair + light final review

48-HOUR PLAN

Pass-first prioritization

- First priority: A2, A4, B1, B2. These are the highest stability sets.
- Second priority: B3 and A1. They have short repeated proofs/theory plus manageable numericals.
- Third priority/insurance: A3 if truss is already familiar; B4 because it reuses friction + impulse.
- Do one 90-minute mini-mock for each section. Do not spend the final night learning a brand-new long truss method from zero.
- Final recall: belt proof, impulse proof, J proof, sector centroid formula, distributed-load locations, friction direction, Pappus formulas.

Healthy study rule

Use focused study blocks with breaks and sleep. A tired 3-hour mock with sign errors is less useful than a rested, timed rehearsal followed by a short error review.

Exam-Hall Strategy: 180 Minutes

The paper gives 6 full questions to answer. Control time by full set.

Stage	Suggested time	Action
Paper scan	5 min	Mark Primary / Backup sets; check figures and data before committing.
6 full questions	~27 min each	Theory/proof + numerical + final answer; move on when time is up.
Final check	13 min	Units, signs, missing subparts, T/C, boxed results, page/question numbering.

Question order

Start with your most certain 35-mark set, not necessarily Q1/Q5. Confidence early prevents time panic. Keep the longest truss/composite M.I. set away from the final 20 minutes.

Per-question writing template

- 1. Rewrite the subpart label clearly: (a), (b), (c).
- 2. Draw the required FBD/sketch before equations.
- 3. State governing equation/formula with symbols.
- 4. Substitute values with units.
- 5. Show one or two key intermediate steps - enough for method marks.
- 6. Box final magnitude + unit + direction / tension-compression / sign meaning.
- 7. Leave a small gap before the next subpart so the examiner can follow the script.

No-Error Checklist

These are the mistakes most likely to destroy otherwise correct work.

- ☐ FBD: every support/contact/cable reaction direction is physically justified.
- ☐ Smooth surface: reaction is normal; cable tension acts along cable.
- ☐ Beam: triangular distributed load location is correct ($L/3$ from heavy end).
- ☐ Beam: pure couple affects moment equilibrium, not force equilibrium.
- ☐ Pulley: same continuous ideal rope has same tension; do not blindly divide W .
- ☐ Truss: final member forces are labelled Tension or Compression.
- ☐ Friction: direction opposes impending/actual relative motion; $F=\mu N$ only when appropriate.
- ☐ Belt: contact angle θ is in radians in $e^{(f\theta)}$.
- ☐ Centroid: holes/cut-outs are negative area.
- ☐ Area M.I.: unit is length^4 ; radius of gyration unit is length.
- ☐ Parallel-axis: transfer distance d is perpendicular distance between parallel axes.
- ☐ Impulse in FPS: use mass $m=W/g$ when momentum/velocity is needed.
- ☐ Work: force opposing displacement gives negative work.
- ☐ Pappus: identify generator and centroid travel radius before applying formula.
- ☐ Final answer includes magnitude, unit and direction/sign interpretation.

Final Memory Sheet

Read this once before entering the exam hall.

If you see...	Immediately think...
FBD / support / smooth contact	Isolate body; replace constraints; normal reaction; cable along rope
Wheel over block	Impending pivot at corner; old contact reaction zero; moment about corner
Beam with UDL/triangle	Equivalent loads → reactions → cut → V/M
Pulley	Same rope same T; draw FBD of each moving pulley
Truss selected members	Whole reactions → section cut ≤3 unknowns → moments strategically
Tall slender body	Calculate slide threshold and tip threshold separately
Wedge	Assume impending direction; rotate contact reaction by friction angle
Composite centroid	Shape table; cut-outs negative
Composite M.I.	Centroid first if needed; standard $\bar{I} + Ad^2$; subtract holes
Impulse	Force-time area = momentum change
Inclined work	Resolve along plane; normal → friction; resultant × displacement
Pappus	Generator × 2π × centroid radius

Recommended final selection mindset

Do not enter the exam with exactly 3 prepared sets per section. Enter with 3 strong + 1 workable backup. The choice rule is your advantage only if the backup is usable.

Made by 2223033

Source & Integrity Note

What this guide is based on - and what it does not invent.

- Primary examination target: Engineering Mechanics Backlog Examination 2023 (CE 1201), 4 pages.
- Regular examination comparison: 2015-2021 merged paper set, 2023 regular paper, and 2024-2025 regular papers supplied in this chat.
- Section A/B set-wise notes used for recurring definitions, formulas, solved patterns and year tags.
- Lecture slides used for centroid, moment of inertia and Pappus-Guldinus concepts/formulas.
- Faires/Analytic Mechanics solution material and class solutions used as supporting solution references where readable.
- 2020 is treated as an exception to the normal four-set distribution as instructed; it is not used to inflate the normal repeat counts.
- Where a scanned figure is ambiguous, this guide emphasizes the governing method rather than inventing a hidden dimension or arrow direction.

Quality principle

For backlog preparation, a correct method is more valuable than a memorized numeric answer. Use the answer checks in this guide to verify practice, but reproduce the FBD, geometry, sign convention and equations in the exam.